Heat-engine efficiency

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 9 times as large. Compare rms speed.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = 1 - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

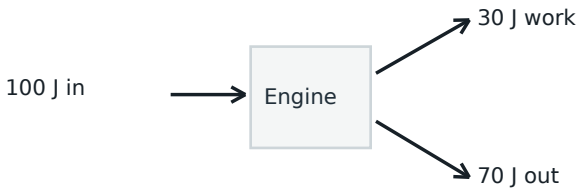
Worked reasoning: A heat engine takes 100 J from a hot source, does 25 J of work and rejects 75 J. Efficiency=25/100=25%.

Your turn: Draw three labeled energy arrows and check conservation.

Marzano check - circle: ① ② ③

03 Represent and interpret

An engine receives 100 J, produces 30 J of work and rejects 70 J. Check the energy balance and find efficiency.



Marzano check - circle: ① ② ③

04 Solve and justify

An engine operates between 600 K and 300 K and receives 800 J per cycle. Find ideal maximum efficiency and maximum work. Actual work is 240 J; find actual efficiency.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Why should a 95% efficient electric motor not be confused with a 95% heat engine converting heat from one source?

Marzano check - circle: 1 2 3

06 Choose a method

At fixed speed, mass becomes 4 times as large while energy-coupling fraction becomes 1/6 as large. Compare coupled energy.

Marzano check - circle: 1 2 3

07 Exit check

A heat engine has Q_{hot}=1500 J and Q_{cold}=1050 J. Find work and efficiency.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

An ideal upper bound is $\eta=1-T_c/T_h$ with kelvin temperatures. For T_h=600 K and T_c=300 K, compare that bound with an engine producing 400 J work from 1000 J input.

Marzano check - circle: 1 2 3